

# Math 3108 Exam 2 Cheat Sheet

## Section 2.2: The Inverse of a Matrix

- **Invertible Matrix:** An  $n \times n$  matrix  $A$  is invertible if there exists an  $n \times n$  matrix  $C$  ( $= A^{-1}$ ) such that  $CA = AC = I_n$ . Only square matrices can be invertible in this sense.
- **Determinant (2x2):** For  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ ,  $\det A = ad - bc$ .
- **Inverse (2x2):** If  $\det A = ad - bc \neq 0$ , then  $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ . If  $\det A = 0$ ,  $A$  is not invertible (singular).
- **Solving  $A\mathbf{x} = \mathbf{b}$ :** If  $A$  is invertible, the unique solution is  $\mathbf{x} = A^{-1}\mathbf{b}$ . (Often used in exam Qs where  $A^{-1}$  is given or calculated first).
- **Properties of Inverses:**
  - $(A^{-1})^{-1} = A$
  - $(AB)^{-1} = B^{-1}A^{-1}$  (Order reverses! Frequently tested).
  - $(A^T)^{-1} = (A^{-1})^T$
- **Finding the inverse (General Method):** Row reduce the augmented matrix  $\begin{bmatrix} A & I \end{bmatrix}$ . If  $A$  is row equivalent to  $I$ , then  $\begin{bmatrix} A & I \end{bmatrix} \sim \begin{bmatrix} I & A^{-1} \end{bmatrix}$ .

## Section 2.3: Characterizations of Invertible Matrices

- **The Invertible Matrix Theorem (IMT):** For a square  $n \times n$  matrix  $A$ , the following are equivalent (all true or all false):
  - $A$  is invertible.
  - $A$  is row equivalent to  $I_n$ .
  - $A$  has  $n$  pivot positions.
  - $A\mathbf{x} = \mathbf{0}$  has only the trivial solution ( $\mathbf{x} = \mathbf{0}$ ).
  - The columns of  $A$  are linearly independent.
  - The linear transformation  $\mathbf{x} \mapsto A\mathbf{x}$  is one-to-one.
  - $A\mathbf{x} = \mathbf{b}$  has a unique solution for each  $\mathbf{b}$  in  $\mathbb{R}^n$ .
  - The columns of  $A$  span  $\mathbb{R}^n$ .
  - The linear transformation  $\mathbf{x} \mapsto A\mathbf{x}$  maps  $\mathbb{R}^n$  onto  $\mathbb{R}^n$ .
  - There exists  $C$  such that  $CA = I$ .
  - There exists  $D$  such that  $AD = I$ .
  - $A^T$  is invertible.
  - $\det A \neq 0$  (from Chapter 3).
  - See also Section 4.5 additions.
- **Key Use:** If you know one condition holds (or fails), you know *all* conditions hold (or fail) for a square matrix. (Many True/False or multiple-choice questions rely on this).

## Section 2.4: Partitioned Matrices

- **Block Multiplication:** Multiply partitioned matrices using row-column rule, treating blocks as elements, *provided the partitions are conformable* (column partition of first matrix matches row partition of second).
  - Example:  $\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} E & F \\ G & H \end{bmatrix} = \begin{bmatrix} AE + BG & AF + BH \\ CE + DG & CF + DH \end{bmatrix}$  (if block sizes match).
- **Inverse of Block Diagonal:**  $\left( \begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix} \right)^{-1} = \begin{bmatrix} A^{-1} & 0 \\ 0 & D^{-1} \end{bmatrix}$  (if  $A, D$  invertible).

## Section 2.5: Matrix Factorizations

- **LU Factorization:** If  $A$  can be row reduced to echelon form  $U$  without row interchanges, then  $A = LU$ .
  - $L$ : Square lower triangular matrix with 1s on the diagonal. Same size as  $A$  if  $A$  is square.
  - $U$ : Echelon form of  $A$ . Same size as  $A$ .
- **Finding LU:**
  1. Reduce  $A$  to echelon form  $U$  using only row replacements ( $R_i \rightarrow R_i - kR_j$ ).
  2. Construct  $L$  by placing the multiplier  $k$  used to eliminate the entry in row  $i$  using row  $j$  into the  $(i, j)$  position of  $L$ . Place 1s on the diagonal and 0s above the diagonal.
- **Solving  $A\mathbf{x} = \mathbf{b}$  using LU:**
  1. Solve  $L\mathbf{y} = \mathbf{b}$  for  $\mathbf{y}$  (using forward substitution).
  2. Solve  $U\mathbf{x} = \mathbf{y}$  for  $\mathbf{x}$  (using backward substitution).

## Section 3.1: Introduction to Determinants

- **Determinant:** A scalar value associated with a square matrix.
  - **Determinant (2x2):**  $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$ .
- **Cofactor:**  $C_{ij} = (-1)^{i+j} \det A_{ij}$ , where  $A_{ij}$  is the submatrix obtained by deleting row  $i$  and column  $j$ .
- **Determinant (nxn):** Cofactor expansion across row  $i$  or down column  $j$ .
  - Row  $i$ :  $\det A = \sum_{j=1}^n a_{ij}C_{ij}$
  - Column  $j$ :  $\det A = \sum_{i=1}^n a_{ij}C_{ij}$
- **Determinant of Triangular Matrix:** Product of the diagonal entries.

## Section 3.2: Properties of Determinants

- **Row Operations:** Let  $A \rightarrow B$ .
  - Row Replacement ( $R_i \rightarrow R_i + kR_j$ ):  $\det B = \det A$ .
  - Row Interchange ( $R_i \leftrightarrow R_j$ ):  $\det B = -\det A$ .
  - Row Scaling ( $R_i \rightarrow kR_i$ ):  $\det B = k \det A$ .
- **Calculating Determinants:** Use row operations (mostly replacements and interchanges) to get to echelon form (triangular). Keep track of determinant changes.  $\det A = (-1)^r \times$  (product of pivots in  $U$ ), where  $r$  is # interchanges. *If scaling is used, factor that out.*
- **Invertibility:**  $A$  is invertible if and only if  $\det A \neq 0$ .
- **Properties:**
  - $\det A^T = \det A$ .
  - $\det(AB) = (\det A)(\det B)$ .
  - $\det(A^k) = (\det A)^k$ .
  - If columns (or rows) are linearly dependent,  $\det A = 0$ .

## Section 3.3: Cramer's Rule, Volume

- **Cramer's Rule:** Solves  $A\mathbf{x} = \mathbf{b}$  for invertible  $A$ .
  - $x_i = \frac{\det A_i(\mathbf{b})}{\det A}$ , where  $A_i(\mathbf{b})$  is  $A$  with column  $i$  replaced by  $\mathbf{b}$ .
- **Inverse using Adjugate:**  $A^{-1} = \frac{1}{\det A} \text{adj } A$ , where  $\text{adj } A = [C_{ji}]$  (matrix of cofactors, transposed).
- **Geometric Interpretation:**
  - Area of parallelogram (in  $\mathbb{R}^2$ ) determined by vectors  $\mathbf{u}, \mathbf{v}$  is  $|\det[\mathbf{u} \ \mathbf{v}]|$ . (Exam problems often give vertices, find vectors from origin).
  - Volume of parallelepiped (in  $\mathbb{R}^3$ ) determined by  $\mathbf{u}, \mathbf{v}, \mathbf{w}$  is  $|\det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}]|$ .

## Section 4.1: Vector Spaces and Subspaces

- **Vector Space:** A set with vector addition and scalar multiplication satisfying 10 axioms (closure, commutativity, associativity, zero vector, additive inverse, etc.). Key examples:  $\mathbb{R}^n$ ,  $\mathbb{P}_n$  (polynomials degree  $\leq n$ ).
- **Subspace:** A subset  $H$  of a vector space  $V$  that is itself a vector space.
- **Subspace Criteria (How to Prove):** Must satisfy ALL three:
  1.  $\mathbf{0} \in H$  (Zero vector of  $V$  is in  $H$ ).
  2. Closed under addition: If  $\mathbf{u}, \mathbf{v} \in H$ , then  $\mathbf{u} + \mathbf{v} \in H$ .
  3. Closed under scalar multiplication: If  $\mathbf{u} \in H$  and  $c$  is a scalar, then  $c\mathbf{u} \in H$ .
- **Span as Subspace:**  $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$  is always a subspace.

## Section 4.2: Null Spaces, Column Spaces, Row Spaces

- **Nul A:** Set of all solutions to  $A\mathbf{x} = \mathbf{0}$ . Subspace of  $\mathbb{R}^n$ .
  - **Basis for Nul A:** Solve  $A\mathbf{x} = \mathbf{0}$ , write solution in parametric vector form. The vectors multiplying the free variables form the basis.
- **Col A:** Set of all linear combinations of the columns of  $A$  (or Range of  $T(\mathbf{x}) = A\mathbf{x}$ ). Subspace of  $\mathbb{R}^m$ .
  - **Basis for Col A:** The pivot columns of the original matrix  $A$ .
- **Row A:** Set of all linear combinations of the rows of  $A$ . Subspace of  $\mathbb{R}^n$ . (Same as  $\text{Col } A^T$ ).
  - **Basis for Row A:** The non-zero rows of the echelon form ( $U$ ) of  $A$ .
- **Kernel of T:** Same as Nul A if  $T(\mathbf{x}) = A\mathbf{x}$ .
- **Range of T:** Same as Col A if  $T(\mathbf{x}) = A\mathbf{x}$ .

## Section 4.3: Linearly Independent Sets and Bases

- **Linearly Independent (LI):** The equation  $c_1\mathbf{v}_1 + \dots + c_p\mathbf{v}_p = \mathbf{0}$  has only the trivial solution ( $c_1 = \dots = c_p = 0$ ).
- **Linearly Dependent (LD):** There exist weights  $c_i$ , *not all zero*, such that  $c_1\mathbf{v}_1 + \dots + c_p\mathbf{v}_p = \mathbf{0}$ . (At least one vector is a linear combo of others).
- **Basis:** A set  $B$  that is **Linearly Independent** AND **Spans** the vector space/subspace.
- **Basis Views (Multiple Choice!):** A basis is a *minimal spanning set* and a *maximal linearly independent set*.
- **Spanning Set Theorem:** Can remove LD vectors from a spanning set without changing the span.
- **Finding Bases:** See methods for Nul A, Col A, Row A in Section 4.2.

## Section 4.4: Coordinate Systems

- **Coordinates:** If  $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$  is a basis for  $V$ , and  $\mathbf{x} = c_1\mathbf{b}_1 + \dots + c_n\mathbf{b}_n$ , then the  $B$ -coordinate vector is  $[\mathbf{x}]_B = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$ . Representation is *unique*.
- **Change-of-Coordinates Matrix:**  $P_B = [\mathbf{b}_1 \ \dots \ \mathbf{b}_n]$ . Then  $\mathbf{x} = P_B[\mathbf{x}]_B$ .
- **Finding  $[\mathbf{x}]_B$ :** Solve the system  $P_B\mathbf{c} = \mathbf{x}$  for the coordinate vector  $\mathbf{c} = [\mathbf{x}]_B$ .
- **Coordinate Mapping:**  $\mathbf{x} \mapsto [\mathbf{x}]_B$  is an isomorphism (preserves structure). Useful for checking LI in spaces like  $\mathbb{P}_n$  by converting to coordinate vectors in  $\mathbb{R}^n$ .

## Section 4.5: The Dimension of a Vector Space

- **Dimension (dim V):** The number of vectors in *any* basis for  $V$ .
- **Rank A:**  $\dim \text{Col } A$  (=  $\dim \text{Row } A$ ) = number of pivot positions.
- **Nullity A:**  $\dim \text{Nul } A$  = number of free variables.
- **Rank Theorem:** For an  $m \times n$  matrix  $A$ : **rank A + nullity A = n** (number of columns). (Very important for finding dimensions).
- **Basis Theorem:** If  $\dim V = n$ :
  - Any LI set of  $n$  vectors in  $V$  is a basis.
  - Any set of  $n$  vectors that spans  $V$  is a basis.
- **IMT Extended (for  $n \times n$  matrix A):** Equivalent conditions include:
  - Columns of  $A$  form a basis for  $\mathbb{R}^n$ .
  - $\text{Col } A = \mathbb{R}^n$ .
  - $\dim \text{Col } A = n$  (rank  $A = n$ ).
  - $\text{Nul } A = \{\mathbf{0}\}$ .
  - $\dim \text{Nul } A = 0$  (nullity  $A = 0$ ).